6.6 Parametric Equations

Name _____ Period ____ Date _____

Pre-Calculus 6.6 (30-minute plan) · Student sheet

I can…

- sketch a curve from parametric equations and show its orientation.
- eliminate the parameter.
- write parametric equations for a graph.

Vocabulary · fill in the missing word

Plane curve — the set of points (x, y) = (f(t), g(t)) for t in an _____

Parametric equations — x = _____ = g(t)

Parameter — the third _____ t

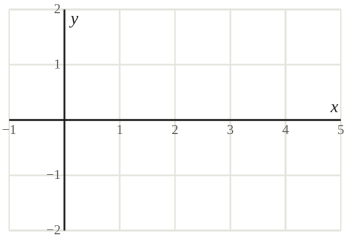
Orientation — the _____ the curve is traced as t increases

Part 1 · Notes and examples (fill in each note, then work the example)

Sketching: make a table of t values, plot (x, y) in order of increasing t, and use arrows to show the _____.

Example 1

Sketch $x = t^2$, $y = \frac{t}{2}$ for $-2 \le t \le 2$. Show the orientation.



Eliminating the parameter: solve one equation for t and _____ into the other.

Example 2

Eliminate the parameter.

a. $x = t - 1$, $y = 2t + 1$

b. $x = t^2$, $y = \frac{t}{2}$

Trig parameters: solve for cos θ and sin θ, then use _____.

Example 3

Eliminate the parameter: $x = 3\cos\theta$, $y = 4\sin\theta$. Identify the curve.

Part 2 · Partner practice

Partner practice

Eliminate the parameter: $x = 2\cos\theta$, $y = 2\sin\theta$.